MASK, ELECTRODE & CONNECTOR GUIDE · XR · XR-16

Electrode Map, Wiring, Power & SWD

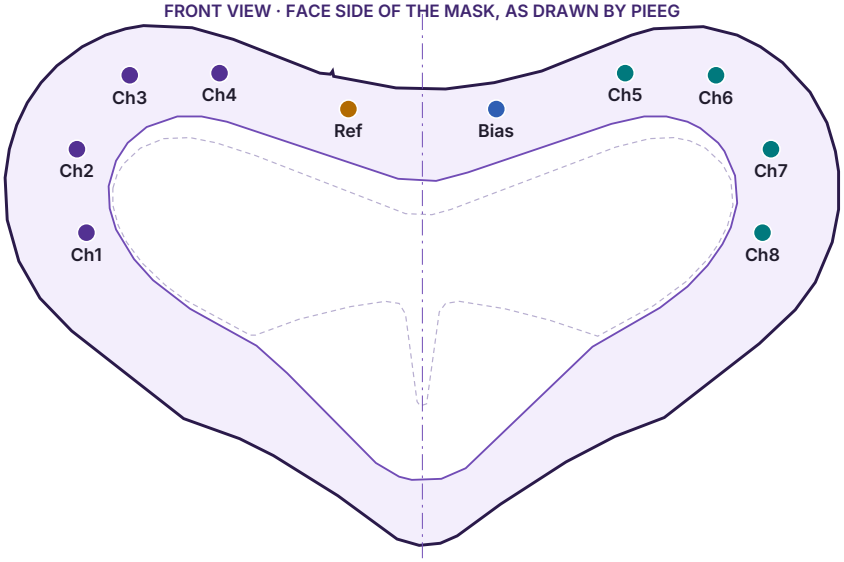
Where the 10 dry electrodes sit on the mask, how they map to the IronBCI-16 electrode header, the USB-C power input and the SWD header used for firmware updates.



10	Ch1–8	Ref	Bias	USB-C
Dry Ag/AgCl electrodes	Signal channels	Centre forehead	Centre forehead	5 V power input

01 ELECTRODE MAP

FRONT VIEW · FACE SIDE OF THE MASK, AS DRAWN BY PIEEG



Ch1–Ch4
left temple and left forehead

Ch5–Ch8
right forehead and right temple

Ref
reference, centre-left forehead

Bias
bias drive, centre-right forehead

10 holes, Ø 1 mm, for flat dry Ag/AgCl electrodes. Exact positions follow the headset model.

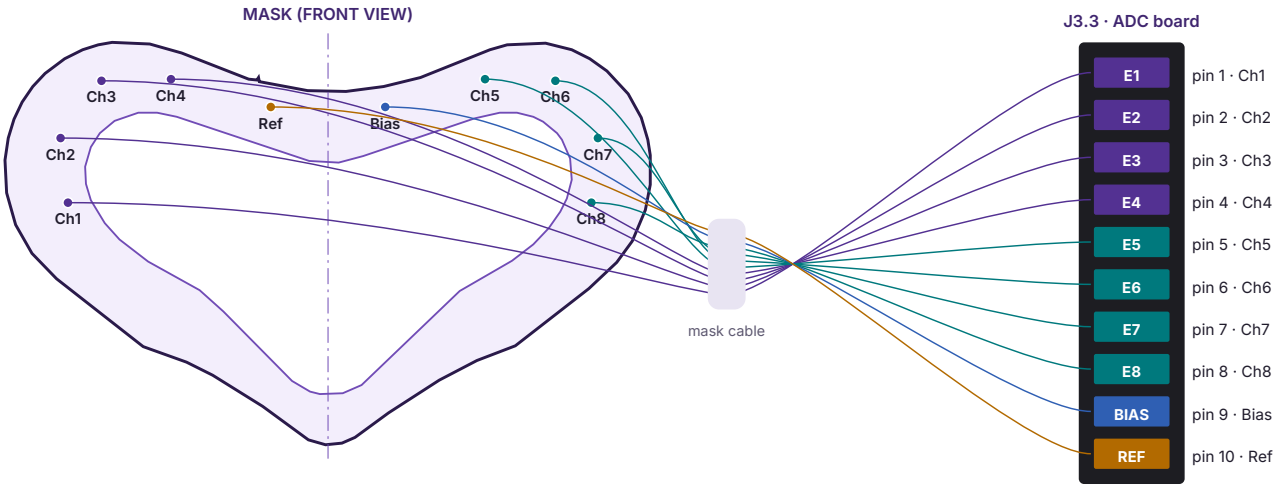
Figure 1 — Electrode positions on the mask (face side, as drawn by PiEEG). Left and right are as drawn.

Position on the mask	Electrodes	Picks up
Left side, lower and upper	Ch1, Ch2	Temple: eye movement, jaw and temple muscles
Left forehead	Ch3, Ch4	Frontal EEG, blinks
Centre forehead	Ref, Bias	Reference and bias
Right forehead	Ch5, Ch6	Frontal EEG, blinks
Right side, upper and lower	Ch7, Ch8	Temple: eye movement, jaw and temple muscles

All electrodes are on the front of the head, so PiEEG XR is strongest at frontal EEG, blinks, eye movements and facial expressions. The eyes-closed alpha test needs electrodes at the back of the head.

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02 MASK TO IRONBCI-16 WIRING



Ch1–Ch8 land on E1–E8 of ADS1299 #1. On PiEEG XR-16 the extra electrodes use E9–E16 on header J3.4.

Figure 2 — The mask cable connects to header J3.3 of the IronBCI-16 ADC board inside the box.

Mask electrode	J3.3 pin	Module input	ADC
Ch1 … Ch8	1 … 8	E1 … E8	ADS1299 #1, channels 1–8
Bias	9	BIAS	Bias drive (common-mode rejection)
Ref	10	REF	Reference for all channels
XR-16 extra electrodes	J3.4, 1 … 8	E16 … E9	ADS1299 #2, channels 9–16

03 POWER INPUT

Item	Detail
Connector	USB-C socket on the BCI box
Voltage	5 V from a power bank or a battery head strap
Cable	**USB-A to USB-C** works most reliably
Inside the box	2 wires to the module's 5 V input

NOTE USB-C to USB-C may not work.

Some USB-C supplies only switch on after USB-C power negotiation, so the box gets no power. Use a USB-A to USB-C cable first.



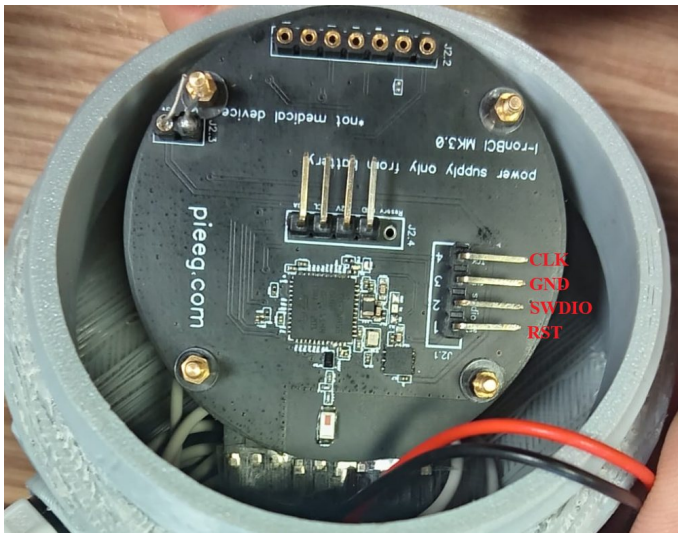
USB-A to USB-C cable.

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04 SWD HEADER (FIRMWARE)

Pin	Signal	ST-Link V2
1	RST	RST (NRST)
2	SWDIO	SWDIO
3	GND	GND
4	SWCLK	SWCLK

The 4-pin angled header is on the MCU board, reached by opening the lid of the box. Pin 1 (RST) is marked on the board; the designator printed next to it depends on the board revision. Flash only with the mask off the face.



SWD header on the board in the BCI box.

05 RELATED DOCUMENTS

XR-DS-001 Product Datasheet	**XR-MD-002** Mechanical & Assembly Drawing	**XR-PCB-003** PCB & Manufacturing Specification
XR-QSG-005 Quick Start & Safety Guide	**XR-SW-006** Bluetooth Interface & Software Guide	

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